HARD · ADVANCED MATH

# Advanced Math

17

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

**Q1** ADVANCED MATH

The area of a rectangular banner is 2,661 square inches. The banner's length $x$, in inches, is 24 inches longer than its width, in inches. Which equation represents this situation?

A. $0 = x^2 - 24x - 2{,}661$

B. $0 = x^2 - 24x + 2{,}661$

C. $0 = x^2 + 24x - 2{,}661$

D. $0 = x^2 + 24x + 2{,}661$

**Q2** ADVANCED MATH

$$ft = 55t - 2t^2$$

The function $f$ is defined by the given equation. The function $g$ is defined by $gt = ft + 3$. Which expression represents the maximum value of $gt$?

A. $3 + \frac{55^2}{2}$

B. $3 + 2\frac{55^2}{4}$

C. $3 - 2\frac{55^2}{4}$

D. $3 - \frac{55^2}{2}$

Q3

ADVANCED MATH

Which of the following functions has(have) a minimum value at -3?

I. $f(x) = -6(3)^{x} - 3$
II. $g(x) = -3(6)^{x}$

**A.** I only

**B.** II only

**C.** I and II

**D.** Neither I nor II

Q4

GRID-IN ADVANCED MATH

| $x$ | $f(x)$ |
|---|---|
| 1 | $a$ |
| 2 | $a^5$ |
| 3 | $a^9$ |

For the exponential function f, the table above shows several values of x and their corresponding values of

$$f(x)$$

, where a is a constant greater than 1. If k is a constant and

$$f(k) = a^{29}$$

, what is the value of k ?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

ADVANCED MATH

$$fx = 1.84^{\frac{x}{4}}$$

The function $f$ is defined by the given equation. The equation can be rewritten as $fx = 1 + \frac{p}{100}^{x}$, where $p$ is a constant. Which of the following is closest to the value of $p$ ?

A. 16

B. 21

C. 46

D. 96

Q6 ADVANCED MATH

$$M = 1{,}800(1.02)^{t}$$

The equation above models the number of members, M, of a gym t years after the gym opens. Of the following, which equation models the number of members of the gym q quarter years after the gym opens?

A. $M = 1{,}800(1.02)^{\frac{q}{4}}$

B. $M = 1{,}800(1.02)^{4q}$

C. $M = 1{,}800(1.005)^{4q}$

D. $M = 1{,}800(1.082)^{q}$

Q7

GRID-IN ADVANCED MATH

The function $f$ is defined by $fx = a^x + b$, where $a$ and $b$ are constants and $a > 0$. In the $xy$-plane, the graph of $y = fx$ has a $y$-intercept at $0, \text{-}25$ and passes through the point $2, 23$. What is the value of $a + b$?

STUDENT-PRODUCED RESPONSE

| | | | |
|---|---|---|---|
| . | / . | / . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8 ADVANCED MATH

$$fx = 2722^{x}$$

The function $f$ is defined by the given equation. If $hx = fx - 4$, which of the following equations defines function $h$?

A. $hx = 172^{x}$

B. $hx = 682^{x}$

C. $hx = 27216^{x}$

D. $hx = 2728^{x}$

Q9 ADVANCED MATH

$$fx = x - 10x + 13$$

The function $f$ is defined by the given equation. For what value of $x$ does $fx$ reach its minimum?

A. -130

B. -13

C. $-\frac{23}{2}$

D. $-\frac{3}{2}$

Q10

ADVANCED MATH

$f(x) = -500x^2 + 25{,}000x$

The revenue $f(x)$, in dollars, that a company receives from sales of a product is given by the function f above, where x is the unit price, in dollars, of the product. The graph of $y = f(x)$ in the xy-plane intersects the x-axis at 0 and a. What does a represent?

A. The revenue, in dollars, when the unit price of the product is $0

B. The unit price, in dollars, of the product that will result in maximum revenue

C. The unit price, in dollars, of the product that will result in a revenue of $0

D. The maximum revenue, in dollars, that the company can make